

## ARTICLE

# Hypo- or hyperarousal? The mechanisms underlying social information processing in autism

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## Abstract

Autistic children tend to show divergent social information processing, and controversies exist regarding the mechanisms underlying this processing. In this article, we summarize and categorize the current theories and evidence related to hypo- and hyperarousal accounts of social information processing in autism. The hypo-arousal account proposes reduced motivation or sensitivity to process social information and is supported by behavioral, eye-tracking, and neuroimaging studies. Recent evidence has challenged this account, highlighting the role of eye gaze in interpreting the hypo-brain activation. The hyperarousal account posits that social context could evoke anxiety and discomfort in autistic children, which could interfere with their social information processing and result in social avoidance. Most research in this field has studied children and adolescents, but some studies have included adults since autism is a lifelong condition. Here, we highlight the importance of considering the heterogeneity of arousal patterns in autistic children when developing individualized supporting protocols.

## KEYWORDS

arousal, autism, eye avoidance, social information processing, social motivation

Autism is a neurodevelopmental condition characterized by difficulties in social interaction and communication, as well as repetitive and routine behaviors. In association with their social difficulties, children with autism

spectrum disorder (ASD) experience specific barriers in social information processing, including difficulties recognizing and discriminating faces, following other people's gazes, understanding emotions and intentions,

**Abbreviations:** ASD, autism spectrum disorder; FFG, fusiform gyrus.

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and understanding and maintaining conversations (e.g., Falck-Ytter et al., 2012; Lord et al., 2020). Social information processing is a building block for social interaction, and its impediment can result in difficulties understanding social cues and relations, and maintaining social relationships. Social information processing could also serve as a promising biomarker of the effectiveness of intervention for autism. However, the social difficulties autistic children face can be explained not only by social information processing but also by nonautistic people with whom they interact, who could have trouble understanding the minds of autistic people (the “double empathy problem”; Milton, 2012, p. 886). Most studies we reviewed focused on children and adolescents, but since autism is a lifelong condition, some studies included adults (see Table S1 for details). However, these studies rarely reported participants' ethnicity and socioeconomic status.

The cognitive mechanisms underlying divergent social information processing in autism have attracted much research attention (Moriuchi et al., 2017). Divergent social information processing in autism has been related to deviations in arousal levels, a state of alertness that could affect one's response to the environment (Cuve et al., 2018). However, researchers have not reached consensus on whether hypo-arousal or hyperarousal underlies this divergent social information processing. The hypo-arousal account, which is represented by social motivation theory (Chevallier, Kohls, et al., 2012) and social brain theory (Pelphrey et al., 2004, 2011), states that autistic children are less sensitive to social stimuli at the behavioral and neural levels than are nonautistic children. They perceive social stimuli as less salient and as having low reward value and show hypo-responsiveness to social stimuli, which accounts for difficulties in social information processing. In contrast, the hyperarousal account states that autistic children are so sensitive to social stimuli that processing these stimuli evokes anxiety and overwhelming feelings, represented by the eye avoidance hypothesis (Tanaka & Sung, 2016) and the intense world theory (Markram & Markram, 2010). To reduce this discomfort and regulate arousal levels, autistic children actively avoid social stimuli, which hinders their social information processing.

Disentangling this contradiction is important not only to understanding more deeply the nature of autism, but also to providing professional support for social

difficulties in individuals with ASD. Current supporting strategies for autism are based primarily on the assumption of autistic children's insensitivity or reduced motivation to process social information (Chevallier, Kohls, et al., 2012). However, if hyperarousal also exists, supporting strategies based on the principles of reinforcement would be inappropriate or inefficient in targeting the core characteristics of autism; instead, strategies should focus on easing the discomfort elicited by social situations under such circumstances.

In this article, we categorize and summarize theories and empirical evidence related to the hypo- and hyperarousal accounts of social information processing in autism (see Table 1). First, we describe the theories and present evidence supporting the hypo-arousal account of social information processing. Then, we discuss challenges to the hypo-arousal account from recent findings and introduce the hyperarousal account of social information processing. Finally, we suggest that researchers consider this contradiction in the empirical and clinical works of autism, which can increase opportunities for more personalized support for autistic children with different arousal patterns.

## THE HYPO-AROUSAL ACCOUNT

In this section, we summarize two important theories representing the hypo-arousal account—social motivation theory (Chevallier, Kohls, et al., 2012) and social brain theory (Pelphrey et al., 2004, 2011). We also describe supporting evidence and recent challenges to the theories.

### Social motivation theory

Social motivation theory proposes that diminished social motivation or reduced social interest could underlie the social and communicative difficulties in autism (Chevallier, Kohls, et al., 2012). Because of divergent reward circuitry in the brains of autistic children, they do not perceive social stimuli to be as rewarding as they are to neurotypical children. As a result, autistic children are less likely to show attentional orientation to social stimuli, seek out and enjoy social interactions, and maintain

**TABLE 1** Theories and evidence of hypo- and hyperarousal in social information processing in autism

|          | Hypo-arousal                                                                                                                                                                                           | Hyperarousal                                                                                                                                                                         |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theories | Social motivation theory (Chevallier, Kohls, et al., 2012)<br>Social brain theory (Pelphrey et al., 2004, 2011)<br>Gaze indifference hypothesis (Moriuchi et al., 2017)                                | Intense world theory (Markram & Markram, 2010)<br>Eye avoidance hypothesis (Tanaka & Sung, 2016)<br>Empathy imbalance hypothesis (Smith, 2009)                                       |
| Evidence | Less response to name<br>Insensitivity to social reputation<br>Reduced eye contact<br>Insensitivity to others' gaze<br>Difficulties in emotion recognition<br>Less activation in social brain networks | Active eye avoidance<br>Increased anxiety under social context<br>Heightened emotional empathy<br>Increased physiological responses to faces<br>Hyper-amygdala activity viewing eyes |



social relationships. This reduced social motivation may deprive them of social learning experiences, which may hinder their development of language and social skills.

Supporting evidence for social motivation theory comes from studies showing autistic children's reduced sensitivity to many kinds of social signals. Attention to social signals is adaptive and often prioritized by neurotypical children (Chevallier, Kohls, et al., 2012). Even in infancy, neurotypical children are very sensitive to social signals (e.g., mother's voice and face, social smiles, and biological motions [e.g., people's movements, including whole-body motion and partial movement of the hands, head, or eye; Shultz et al., 2018]). However, autistic children are often less responsive to social stimuli or signals, such as responding to their own names, a developmental milestone typically achieved in infancy: In one study, 6- to 24-month-olds later diagnosed with autism were more likely to fail at responding to names or need more prompts to respond to names than were neurotypical infants of the same age (Miller et al., 2017). Studies involving the still-face paradigm provide further supporting evidence: 20-week-olds at increased likelihood of autism were less affected by the changes in social context (i.e., their mother's change in facial expressions) than were neurotypical infants (Yirmiya et al., 2006).

Joint attention is another area of social attention in which autistic individuals have difficulty: Many studies have revealed autistic children's difficulty following others' pointing and gazes, and initiating joint attention (e.g., a study of children whose average age was 6 years; Falck-Ytter et al., 2012). Moreover, in a review of studies of 6- to 41-year-olds, autistic children and adults also showed insensitivity to more complex social cues, such as others' emotions (Uljarevic & Hamilton, 2013). In a study of children whose age averaged 10 years, this was characterized by difficulty in emotion recognition (Vogindroukas et al., 2014); in another study, of adolescents averaging 15 years old, this was particularly so in recognizing sadness, disgust, anger, and surprise (Smith et al., 2010). Diminished social interest in autism was also associated with reduced reputation management in a study of autistic adolescents who were, on average, 13 years old (Chevallier, Molesworth, et al., 2012), although that finding has been challenged by recent studies suggesting that school-age autistic children have intact reputation management (e.g., a study of children whose age averaged 11 years; Cage et al., 2016) and self-representation (e.g., a study of children whose age averaged 10 years; Scheeren et al., 2016).

## Social brain theory

Neuroimaging studies provide more evidence of hypo-arousal in social information processing in autistic

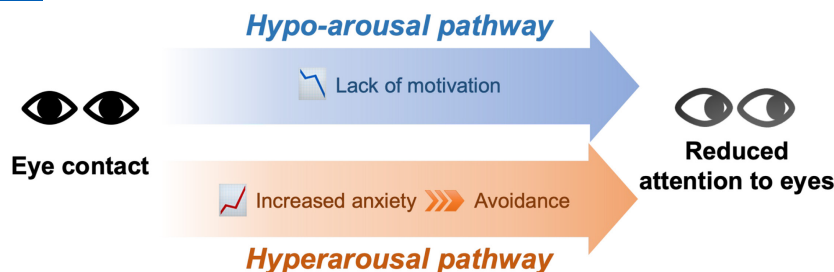
individuals. Social brain theory proposes that dysfunction in social brain networks, including the amygdala, the superior temporal sulcus, and the fusiform gyrus (FFG), underlies social difficulties in autism (Pelphrey et al., 2004, 2011). First, the amygdala plays an important role in emotion processing and guiding attention to biologically and socially relevant stimuli (e.g., the eyes; Gamer & Büchel, 2009). Second, compared to neurotypical individuals, autistic children had reduced activation in the posterior superior temporal sulcus when viewing biological motions and understanding goal-directed actions and intentions (Pelphrey et al., 2004, 2011). Third, hypo-activation of the FFG and the amygdala could underlie divergent face perception in autistic adults (Perlman et al., 2010; Pierce, 2001).

## Challenges to the hypo-arousal account

Social brain theory posits that dysfunction in the social brain network accounts for reduced social attention (Pelphrey et al., 2004, 2011). However, such hypo-responsivity could be a consequence rather than a cause of reduced social attention in autistic children and adults (as seen in a study of autistic individuals whose age averaged 23 years, Hadjikhani et al., 2017, and a study of individuals whose age averaged 16 years, Tottenham et al., 2014). Hypo-activity in certain brain areas may be due to autistic individuals' attentional avoidance of social stimuli. Many empirical studies based on eye-tracking and neuroimaging techniques have supported this view.

For instance, in a study of autistic children and adults whose age averaged 15 years, eye contact correlated positively with increased amygdala response to faces (Dalton et al., 2005). In a subsequent study, decreased eye contact in autism reflected active avoidance of the eye region rather than autistic adults' hypo-sensitivity to the eyes (Kliemann et al., 2010). Furthermore, cueing or constraining autistic children's and adults' gaze to the eye regions (i.e., forced eye contact in autism) increased both cortical (e.g., FFG) and subcortical (e.g., amygdala) brain activity (Hadjikhani et al., 2017; Kliemann et al., 2012; Perlman et al., 2010). Researchers should consider eye gaze when interpreting amygdala hypo-activation in autism because divergent eye gaze was a confounding factor for the findings of amygdala hypo-activation in autism, and thus should be controlled in brain imaging studies of autistic individuals (Stuart et al., 2022).

Social motivation theory (Chevallier, Kohls, et al., 2012) has also been challenged by alternative explanations based on empirical evidence and the testimony of autistic individuals (Jaswal & Akhtar, 2019). These challenges to the hypo-arousal account call for considering intense emotional and physiological reactivity elicited



**FIGURE 1** Hypo- and hyperarousal pathways (with reduced attention to eyes as an example). *Note:* In the hypo-arousal pathway, autistic children show diminished social motivation and social interest, resulting in their reduced attention to eyes (social motivation theory; Chevallier, Kohls, et al., 2012). In the hyperarousal pathway, eye contact would evoke hyperarousal and overwhelming anxiety in autistic children, who then avoid looking at the eyes to reduce tension and discomfort (eye avoidance hypothesis; Tanaka & Sung, 2016).

in social situations and their impact on social information processing, which is highlighted in the hyperarousal account.

## THE HYPERAROUSAL ACCOUNT

Autistic children tend to exhibit more frequent anxiety than neurotypical children, especially in social contexts. This excessive anxiety and accompanying worry and fear regarding social situations may be a barrier to processing social information, engaging in social activities, and establishing social relationships. The hyperarousal account emphasizes the role of emotional and physiological responses in social information processing in autism, rather than the reduced motivation and sensitivity suggested by the hypo-arousal account. Next, we summarize the relevant theories of the hyperarousal account to explain the difficulties in social information processing in autism and their supporting evidence.

### The eye avoidance hypothesis

Information from the eyes is assumed to be vital in social interactions. However, in a review of studies of 4-month-olds to 40-year-olds, autistic children and adults showed reduced attention to the eyes relative to neurotypical children and adults (Frazier et al., 2017). This divergency can be detected in infants later diagnosed with ASD, and persists into childhood and adulthood (e.g., Frazier et al., 2017 [in a meta-analysis covering all ages]; Jones & Klin, 2013 [studying 2- to 24-year-olds]; and Moriuchi et al., 2017 [studying children whose average age was 25 months]). The gaze indifference hypothesis attributes this reduced eye looking to autistic individuals' insensitivity to others' eyes (Moriuchi et al., 2017). Attenuated eye looking in autism is in line with social motivation theory, which suggests that eye gaze and social information are insufficiently rewarding for autistic children (Chevallier, Kohls, et al., 2012).

In contrast, the eye avoidance hypothesis proposes that autistic children perceive eye contact as socially threatening and thus actively avoid looking at others' eyes to relieve their uncomfortable feelings (Tanaka & Sung, 2016). On the behavioral level, 40-month-old autistic children avoided the eyes of threatening faces (e.g., angry faces) more than the eyes of nonthreatening faces (e.g., neutral, happy, sad faces; e.g., Wang et al., 2018), and autistic children and adults who rated faces as more threatening made fewer eye movements toward the eye region of a neutral face during free viewing (Tottenham et al., 2014). On the physiological level, 10-year-old autistic children had increased physiological responses to faces with a direct gaze than to faces with an averted gaze (Kylliäinen & Hietanen, 2006). On the neural level, autistic children and adults experienced hyper-amygdala activity when they looked at people's eyes, especially when the people's faces had a direct gaze and fearful, neutral, or ambiguous expressions (for a review, see Stuart et al., 2022). The eye avoidance hypothesis challenges social motivation theory and emphasizes the role of affective arousal. That is, instead of reduced motivation, some autistic children might avoid social stimuli to reduce their tension and discomfort (see Figure 1).

### The empathy imbalance hypothesis

Autistic children are described as being indifferent to other people's distress and less likely to offer comfort to others in distress. Based on findings from self- and parent-report scales and clinical observations, autistic adolescents (mean age of 15 years) tend to show difficulties in cognitive empathy and empathic concern (Vilas et al., 2021). However, in studies using skin conductance, pupillometry, and brain activations to measure emotional empathy, an unconscious and automatic process, autistic adults showed more complex patterns of empathy (e.g., Gu et al., 2015). According to the empathy imbalance hypothesis (Smith, 2009), autistic children have less cognitive perspective taking but more emotional arousal. In support of this



hypothesis, in one study, compared to neurotypical adults, autistic adults tended to display intact or even heightened emotional empathy, which is characterized by hyperactivation in the somatosensory cortex and the anterior insular cortex, as well as enhanced skin conductance response when viewing others' pain (e.g., Gu et al., 2015).

## Intense world theory

The intense world theory of autism, based on the valproate rat model, considers enhanced brain functioning the core of autism (Markram & Markram, 2010). This hyperfunctioning originates from hyperreactivity and hyperplasticity of microcircuits on the neurobiological level, which leads to hypersensitivity, hyperfear, and hyperemotionality. These deviations on perceptual and cognitive levels would cause autistic children to perceive the world as intense and aversive, leading them to avoid and even withdraw from stimuli with high arousal levels, such as social scenes.

## ISSUES IN AUTISM RESEARCH AND PRACTICE

### Limitations of the current theories

Overall, the theories on hypo- or hyperarousal patterns in autism provide important insights into the underlying mechanisms of social information processing. However, we also acknowledge the limitations of some of the theories in lacking direct evidence and deriving testable hypotheses. For example, social motivation theory could explain the relevant empirical findings (e.g., reduced eye contact), but alternative hypotheses could also explain these findings (e.g., the eye avoidance hypothesis).

Another issue with the current theories is their emphasis on gaze in social communication, which is only one potential cue to reflect others' intentions. Autistic children may use other social cues to maintain social engagement, such as physical proximity, body position, head orientation, manual gestures, and vocalizations (Akhtar & Gernsbacher, 2008; Akhtar & Jaswal, 2020). Indeed, in one study, 36-month-old neurotypical and autistic children mainly followed parents' hands rather than their gaze to establish joint attention (Yurkovic-Harding et al., 2022). However, these aspects of communication in autism have not been widely addressed, warranting more research attention.

As mentioned earlier, the studies we reviewed rarely reported participants' demographic information (e.g., ethnicity, socioeconomic status). Limited demographic data prevent researchers from examining the impact of this information on autism and clinical practice. Moreover, these studies are mostly from the United States and Europe (see Table S1 for details), which may

affect the generalizability of their conclusions to non-Western societies. Researchers in this field should provide more demographic information and include more diverse samples.

## Embracing heterogeneity

Although hypo- and hyperarousal appear to be contradictory because they hypothesize the involvement of different cognitive and neural mechanisms in autism, we suggest that these accounts may coexist. As a result of individual differences in autistic people, some autistic children may show hyperarousal to social information, whereas others could show the exact opposite (see Cuve et al., 2018, for more information on the heterogeneity of arousal patterns in emotion recognition in autism). Some autistic children might be less sensitive to the meaning of the eyes, and thus less oriented to this socially salient region, showing hypo-arousal patterns. But attention to the eyes could cause hyperarousal in other autistic children, resulting in eye avoidance as a strategy to alleviate overwhelming feelings.

Co-occurring conditions of autism may also play important roles in the heterogeneity of arousal. For example, anxiety disorder, which may be related to hyperarousal to social information, frequently coexists with autism (Lord et al., 2020); autistic children with a co-occurring anxiety disorder may be more likely to show hyperarousal patterns than those without such a disorder. Researchers should test the effects of co-occurring conditions of autism on patterns of arousal. In addition, the biased sampling in studies in this field, most of which have examined male autistic participants without intellectual disability, hinders a comprehensive understanding of the heterogeneity of arousal patterns in autism.<sup>1</sup> Researchers should include autistic children with different co-occurring conditions as well as underrepresented autistic children, autistic children with intellectual disabilities, and female autistic individuals.

## Arousal patterns across context and age

At the individual level, we speculate that the same autistic person could show patterns of both hyper- and hypo-arousal, depending on the context. For example, autistic children may experience hypo-arousal, which keeps them from actively initiating or joining social activities; however, once they are involved, they may switch from the hypo-arousal to the hyperarousal state, which makes their feelings much more intensive than those of neurotypical children. Researchers should examine different contexts to test their effects on the variance of arousal patterns in autism.

Arousal patterns at the individual level could also vary across ages. Investigations of developmental

processes could shed light on the debate between the hypo- and hyperarousal accounts. Autistic children may experience a hypo-hyper or a hyper-hypo developmental trajectory of their arousal patterns. For example, some researchers have suggested that indifference to gaze may precede avoidance of gaze (Moriuchi et al., 2017). In fact, the indifference hypothesis has been supported by studies of toddlers (Moriuchi et al., 2017), whereas the avoidance hypothesis has been supported by studies of older children, adolescents, and adults (Dalton et al., 2005; Kliemann et al., 2010, 2012; Kylliäinen & Hietanen, 2006). Longitudinal studies are necessary to further understand the developmental changes of arousal patterns and whether arousal patterns have downstream consequences for social abilities in autism. Recognizing the issues of heterogeneity, developmental changes, and contextual variation of arousal patterns could help autism researchers revisit conflicting theories and findings, and inspire their investigations.

## The impact of arousal on daily functioning

In autism, arousal plays an important role not only in social information processing, but also in other aspects of daily functioning. For example, restricted and repetitive behaviors are considered a strategy for regulating arousal levels in 10-year-old autistic children (Lidstone et al., 2014). Hyperaroused children use these behaviors to block excess signal inputs to reduce their arousal level, while hypo-aroused children use them to maintain their sensory stimulations to regulate the underactivation of their automatic nervous systems. However, research on the role of arousal in other aspects of daily functioning (e.g., executive functioning) is limited, which calls for systematic investigations.

## The role of sensory processing in autism

Atypical sensory processing is commonly reported in autism, with about 45% to 95% of autistic children showing atypical sensory behaviors (Ben-Sasson et al., 2009). Divergent sensory responsivity could alter arousal levels in response to social stimuli in autism. Autistic children with sensory over-responsivity may shift their attention to or allocate many cognitive resources to manage sensory stimuli, which may interfere with their social information processing (Green et al., 2018; see also Thye et al., 2018). For example, autistic children with sensory over-responsivity showed decreased activity in the auditory language region of the brain in a social cognition task with the presence of a tactile distractor, as seen in a study of children whose age averaged 14 years (Green et al., 2018). In contrast, under-responsivity to sensory stimuli, including social stimuli, could result in delayed or lack of responses, similar to hypo-arousal (Thye

et al., 2018). Apart from divergent responsivity, variations in basic sensory processing could also result in altered arousal patterns in response to social stimuli. For example, face processing in social interaction requires rapid and complex processing of visual information (aside from social information), which is hindered in autism (Thye et al., 2018). We need more evidence to understand more fully the role of sensory processing in arousal patterns in response to social stimuli in autism.

## Clinical considerations

Many professional support systems for autism rely on the hypo-arousal account that autistic children are less sensitive or motivated to process social information or engage in social activities. Some supporting strategies are designed to strengthen children's motivation to maintain social relationships (e.g., the Early Start Denver Model) or to improve their motivation for verbal communication and social contact (e.g., Pivotal Response Treatment). Other evidence-based support strategies (e.g., the Preschool Autism Communication Trial, Joint Attention Symbolic Play, and Engagement and Regulation) involve encouraging autistic children to increase social interactions and communication (for a review, see French & Kennedy, 2018).

However, many autistic children can instead be hyper-responsive to social signals. Therefore, support systems based on the hypo-arousal account may not be effective for all cases of autism and may even exacerbate difficulties under certain circumstances. For these cases, mindfulness-based intervention might be more appropriate to reduce anxiety and emotion regulation difficulties that arise from hyper-reactivity to social signals. Using systematic desensitization paradigms might also be effective for dealing with specific phobias and hypersensitivity to auditory stimuli. Research based on firsthand reports of autistic teenagers and adults reveals a diversity of experiences with social interactions (Trevisan et al., 2017), highlighting the importance of considering autistic children's personal perspectives in individualized clinical practices.

## CONCLUSION

Divergent social information processing in autism could be explained by hypo- or hyperarousal responses to social stimuli. Each account has gained support from studies, and these findings, although contradictory, provide important insights for theoretical issues and clinical considerations, such as heterogeneity, developmental trajectories, and the role of sensory processing divergencies in arousal patterns in autism. Existing empirical evidence and recommendations should be considered with caution in terms of the arousal patterns underlying atypical behaviors



in autism. Moreover, considering the heterogeneity in arousal patterns in autism, current support strategies may not be appropriate for all autistic children. In this article, we have highlighted the importance of developing individualized supporting protocols for hypoaroused and hyperaroused autistic children.

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## ENDNOTES

<sup>1</sup> Exposure to valproic acid during pregnancy is related to an increased likelihood of being diagnosed with ASD in offspring (Markram & Markram, 2010).

<sup>2</sup> Since autism is more than four times more common in boys and girls, recruiting autistic girls to have a gender-balanced sample is difficult.

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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